Module 3: Probability and Distributions

307312 Applied Statistics | University of Petra

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Module 3: Probability and Distributions

Five lessons | From event rules to probability models

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Learning sequence

Module 2 described recorded data. This module defines uncertainty about a selection or future outcome, then calculates probabilities under explicit models. Finish the descriptive-statistics review first.

1. **Probability rules:** outcomes, events, complements, intersections, and unions.
2. **Conditional probability:** reference groups, independence, multiplication, and Bayes' theorem.
3. **Discrete random variables:** a probability mass function, cumulative probabilities, expected value, and variance.
4. **Binomial and Poisson models:** assumptions, count probabilities, exposure, and simulation.
5. **Continuous distributions:** uniform areas, normal tails, standardization, and quantiles.

Lessons and practice

Lesson	Main question	Practice
1. Probability rules	What event and selection mechanism define the probability?	Lesson 1 practice
2. Conditional probability	How does new information change the reference group?	Lesson 2 practice
3. Discrete random variables	What values can occur, with what probabilities?	Lesson 3 practice
4. Binomial and Poisson models	Which count mechanism and event apply?	Lesson 4 practice
5. Continuous distributions	Which area or cutoff answers the question?	Lesson 5 practice

Finish with the integrated probability and distributions review. Separate instructor guides cover each practice and the review.

Cases, inputs, and readings

Use this guide to connect each case to its input type and selected textbook reading.

Case	Input used	Main 8th-edition reference
Restaurant selection	Existing checked 300-row CSV	Tables 2.9–2.10, pp.65–66; probability rules in §§4.1–4.4
Two suppliers	Stated shares and good/bad probabilities	§4.5, pp.201–205
DiCarlo Motors	Six published daily-sales frequency cells	§§5.2–5.3, pp.222–228
Martin Clothing Store	Three customers, purchase probability 0.30	§5.5, especially p.244
Arrivals	Mean 10 per 15 minutes and stated process assumptions	§5.6, pp.252–255
Flight duration	Uniform bounds 120–140 minutes	§6.1, pp.275–277
Grear tires	Normal mean 36,500 miles and SD 5,000 miles	§6.2, especially pp.286–290

The Restaurant CSV retains its documented provenance and verification limits. All other inputs are explicitly identified printed aggregate values or model parameters entered in the lesson code. They are not new individual-observation datasets. Artificial simulation is labeled as such.

Read each case's English introduction followed by its equivalent Arabic narration. Statistical explanations and interpretation tasks remain in English.

Scope and next step

The selected sequence includes five lessons, five practices, and an integrated review. It is not exhaustive coverage of Chapters 4–6: detailed permutation exercises, bivariate portfolio distributions, hypergeometric calculations, and the exponential distribution remain possible extensions.

Module 4: Sampling and estimation is the next topic; its four lessons, practice, and review are available. We will distinguish the distribution of an individual observation from the sampling distribution of a statistic.

Return to the course home.

Module 3: Probability Rules

Lesson 1 | Define the outcome before calculating its probability

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From description to probability

Module 2 summarized recorded observations. Probability describes uncertainty about an outcome. We begin with a deliberately precise random selection from the same Restaurant data, then use probability models for other textbook settings.

By the end, you should be able to:

- Define an experiment, a sample space, an outcome, and an event.
- Distinguish equally likely selection, historical relative frequencies, and judgment-based probability.
- Calculate complements, intersections, and unions without double counting.
- State the reference group and the limits of the answer.

Reading: main textbook, §§4.1–4.3, printed pp.173–190. Restaurant context and counts: Tables 2.9–2.10, pp.65–66. The selection questions below are course adaptations of that checked textbook example.

Business case: select one restaurant record

An analyst has the book's historical data for **300 Los Angeles restaurants**, with quality ratings and meal prices in US dollars. Suppose a computer selects **one of these 300 records uniformly at random**. What is the probability that the selected restaurant is rated Very Good, has a meal price from USD 20 up to but not including USD 30, or satisfies both conditions? The random outcome is one recorded restaurant; the questions do not predict the current restaurant market.

لدى محلّل بيانات الكتاب التاريخية عن **ثلاثمئة مطعم في لوس أنجلوس**، وتشمل تقييمات الجودة وأسعار الوجبات بالدولار الأمريكي. لنفترض أن الحاسوب يختار **سجلًا واحدًا من هذه السجلات الثلاثمئة عشوائيًا باحتمالات متساوية**. ما احتمال أن يكون تقييم المطعم المختار Very Good، أو أن يكون سعر وجبته من 20 دولارًا إلى أقل من 30 دولارًا، أو أن يتحقق الشرطان معًا؟ النتيجة العشوائية هي مطعم واحد مسجّل، ولا تتنبأ هذه الأسئلة بسوق المطاعم الحالي.

Outcomes and events

An **experiment** is a process with well-defined possible outcomes. Here, the experiment selects one record. The **sample space**, S, contains all 300 restaurant identifiers. One identifier is one **outcome**. An **event** is a set of outcomes satisfying a stated condition.

Define:

- A: the selected restaurant is rated Very Good.
- B: its meal price satisfies $20 \leq$ price < 30 USD.

The half-open price interval includes USD 20 and excludes USD 30. The possible outcomes are the 300 records, rather than the three rating categories: categories contain different numbers of records and are not equally likely.

Expression	Meaning	R operation
A^c	Not A	`!A`
$A \cap B$	Both A and B	`A & B`
$A \cup B$	A or B, including both	`A | B`
S	Any possible outcome	All 300 records

Assign a probability for a reason

A probability lies between 0 and 1; the probabilities of all mutually exclusive, exhaustive outcomes sum to 1.

Basis	Why it can be used	Important qualification
Equally likely outcomes	The selection mechanism gives each outcome the same chance	Count favorable outcomes divided by all outcomes only when equal likelihood is justified
Relative frequency	Past observations describe how often an event occurred	Using that frequency for a future outcome assumes the past is relevant
Subjective assessment	Relevant information and professional judgment support an assessment	Document the basis; precision alone does not make the assessment reliable

For the specified uniform selection, each record has probability $1/300$. Therefore, an event containing k records has probability $k/300$. The resulting probabilities are **exact for this finite selection experiment**. Generalizing them to new restaurants would need additional evidence about sampling and relevance.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# nrow(): returns the number of rows in a table or matrix.
# sum() counts TRUE event indicators; the denominator is all 300 equally selectable
 ↪  records.
# kable() displays counts and probabilities without rounding the underlying calculations.
events <- data.frame(Event = c("A: Very Good", "B: USD 20 to <30", "A and B"),
                     Count = c(sum(A), sum(B), sum(A & B)))
events$Probability <- events$Count / nrow(r)
kable(events, digits = 4)
```

Event	Count	Probability
A: Very Good	150	0.5000
B: USD 20 to <30	118	0.3933
A and B	64	0.2133

Interpretation: $P(A) = 150/300 = 0.50$, $P(B) = 118/300 \approx 0.3933$, and $P(A \cap B) = 64/300 \approx 0.2133$.

Complement and addition rules

The **complement** contains every outcome outside the event:

$$P(A^c) = 1 - P(A).$$

Because A and B overlap, adding their probabilities counts the overlap twice. Subtract it once:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

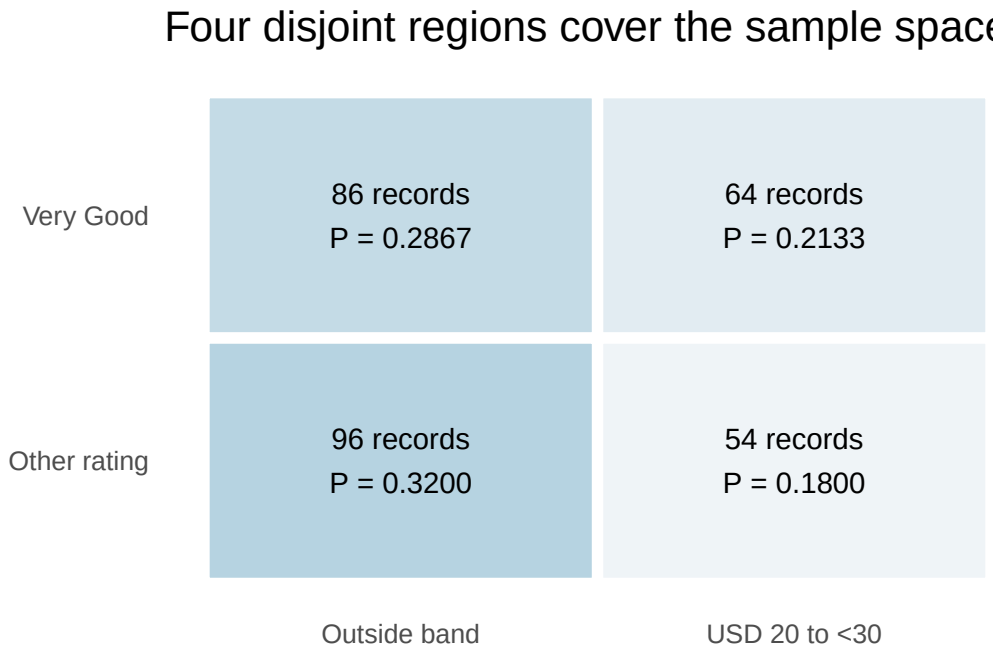
```
# c(): combines values into a vector, converting them to a common type when needed.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# names(): reads or assigns names to vector elements or table columns.
# unname(): removes names or dimension names without changing the underlying values.
# !, &, and | form the complement, intersection, and inclusive union of event indicators.
# Check the addition rule using unrounded values; mean() is the fraction of TRUE records.
results <- c("Not Very Good" = mean(!A),
             "Very Good or USD 20 to <30" = mean(A | B),
             "Very Good but outside that price band" = mean(A & !B))
stopifnot(abs(mean(A | B) - (mean(A) + mean(B) - mean(A & B))) < 1e-12)
kable(data.frame(Event = names(results), Probability = unname(results)), digits = 4)
```

Event	Probability
Not Very Good	0.5000
Very Good or USD 20 to <30	0.6800
Very Good but outside that price band	0.2867

The union contains $150 + 118 - 64 = 204$ records, so its probability is **0.68**. "Or" includes restaurants that satisfy both conditions. "Exactly one of the conditions" is a different event: exclude the overlap from that union.

Two events are **mutually exclusive** when they cannot occur together. For example, Very Good and Excellent are mutually exclusive ratings for one selected restaurant. In that situation $P(A \cap B) = 0$, so the addition rule has no overlap term. Our rating and price events are not mutually exclusive.

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# rep(): repeats values; times repeats a vector and each repeats individual elements.
# c(): combines values into a vector, converting them to a common type when needed.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# paste0(): joins text with no separator; collapse can combine a vector into one string.
# sprintf(): formats text using placeholders such as %.2f for two decimal places.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_text(): draws labels from mapped data; position and hjust/vjust control placement.
# scale_fill_gradient(): maps numeric values to a continuous fill-color gradient.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# theme(): customizes non-data plot elements such as legends, grids, and axis text.
# element_blank(): removes the selected non-data plot element.
# The four disjoint cells partition all 300 records; their counts must add to the total.
# geom_tile() shows region membership; labels carry the counts so color is not essential.
regions <- data.frame(Rating = rep(c("Very Good", "Other rating"), each = 2),
                      Price = rep(c("USD 20 to <30", "Outside band"), 2),
                      Count = c(sum(A & B), sum(A & !B), sum(!A & B), sum(!A & !B)))
regions$Label <- paste0(regions$Count, " records\nP = ", sprintf("%.4f", regions$Count /
 ↪  300))
stopifnot(sum(regions$Count) == 300)
ggplot(regions, aes(Price, Rating, fill = Count)) +
  geom_tile(color = "white", linewidth = 2) +
  geom_text(aes(label = Label), size = 4) +
  scale_fill_gradient(low = "#EFF4F7", high = "#B6D3E1", guide = "none") +
  labs(x = NULL, y = NULL, title = "Four disjoint regions cover the sample space") +
  theme(panel.grid = element_blank())
```



Original R illustration from the checked Restaurant CSV. Cells represent event regions, not geographic locations or additional observations.

Counting possibilities is not assigning probabilities

For two distinct restaurant records selected without replacement, there are 300×299 **ordered** pairs. If order is irrelevant, each pair has been counted twice, giving $300 \times 299/2 = 44{,}850$ **unordered** pairs. More generally, `choose(N, n)` counts selections of n objects from N when order does not matter.

```r
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# choose(300, 2) counts pairs without order; 300*299 counts first-then-second selections.
# Counts become probabilities only after a selection mechanism and an event are specified.
kable(data.frame(Selection = c("Ordered pair", "Unordered pair"),
                 Possibilities = c(300 * 299, choose(300, 2))))
```

Selection	Possibilities
Ordered pair	89700
Unordered pair	44850

Before moving on

For every answer, state **the experiment, the event, the denominator, and the interpretation**. A probability of 0.68 means 68% under repeated selections using this mechanism; it does not guarantee exactly 68 matching records in every 100 selections.

Complete Lesson 1 practice, then continue to conditional probability.

Sources and limits

Main reference: Anderson et al. (2020), Essentials of Modern Business Statistics with Microsoft Excel, 8th ed., §§4.1–4.3 and Tables 2.9–2.10. Probability questions and charts are original course adaptations. Restaurant CSV and data guide: 300 rows; first 10 displayed records and all 12 printed crosstab cells match. Remaining individual rows were not printed for comparison. The probabilities above depend only on checked aggregate counts.

Module 3: Probability Rules — Practice

Lesson 1 | Calculate, justify, and interpret

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Preparation

Read Lesson 1. Use uniform selection of one record from the checked Restaurant CSV. Let A mean Very Good and B mean a price from USD 20 to less than USD 30. The full business introduction is in the lesson.

```
# read.csv() imports the checked teaching file; the analysis leaves its observations
 ↪  unchanged.
# >=20 and <30 define a half-open price band, so USD 30 is excluded.
r <- read.csv("data/module02/restaurant_ratings.csv")
A <- r$quality_rating == "Very Good"
B <- r$meal_price_usd >= 20 & r$meal_price_usd < 30
```

1. Define the experiment

State the sample space, one outcome, and the events A and B. Explain why restaurant identifiers are categorical labels even though R may import them as numbers. Are the three quality categories equally likely in this experiment?

2. Count event regions

Find the counts for both conditions, only A, only B, and neither condition. Check that they total 300. Create a labeled four-cell table or chart.

3. Calculate and interpret

Calculate $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, $P(A^c)$, and the probability of **exactly one** of A and B. Show the denominator and interpret the difference between “or” and “exactly one.”

4. Explain two errors

1. A student adds $P(A) + P(B)$ to calculate the union. Identify the double-counted records and correct the result.
2. Another student says the result proves the proportion of all current Los Angeles restaurants with those characteristics. Explain the limitation.

5. Count pairs

How many ordered pairs of distinct records are possible without replacement? How many unordered pairs? Explain why a count of possibilities alone does not identify an event probability.

Submit a short Quarto report with the calculations, one display, and interpretations. Return to the lesson.

Module 3: Conditional Probability

Lesson 2 | Change the reference group and update a probability

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What we will learn

A condition changes the outcomes still under consideration. We connect the conditional percentages in Module 2 to probability notation, test independence within a specified experiment, and use Bayes' theorem to reverse a conditional question.

Reading: main textbook, §4.4, printed pp.193–198, and §4.5, pp.201–205. Complete Lesson 1 first.

Business case: the same restaurants, a new condition

One record is selected uniformly from the **300 historical Los Angeles restaurants**. The analyst now learns that its meal price is from **USD 20 up to but not including USD 30**. Among records meeting this condition, what is the probability of a Very Good rating? How does that answer differ from asking for the price band among Very Good restaurants?

يُختار سجل واحد باحتمالات متساوية من بيانات **ثلاثمئة مطعم تاريخي في لوس أنجلوس**. يعرف المحلّل الآن أن سعر الوجبة في المطعم المختار **من 20 دولارًا إلى أقل من 30 دولارًا أمريكيًا**. داخل السجلات التي تحقق هذا الشرط، ما احتمال أن يكون التقييم Very Good؟ وكيف تختلف الإجابة عن احتمال الانتماء إلى الفئة السعرية نفسها بين المطاعم ذات التقييم Very Good؟

```
# library(): loads a package so its functions can be used in this session.
# read.csv(): reads a comma-separated file into a data frame; its first row supplies
 ↪  column names by default.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# length(): counts elements in a vector or list.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# theme_set(): sets the default theme for subsequent ggplot charts in this R session.
# theme_minimal(): uses a minimal chart style; base_size controls the base text size in
 ↪  points.
# Recreate the same events so conditioning changes the reference group rather than the
 ↪  data.
# mean(A[B]) averages A only within records satisfying B; this requires at least one B
 ↪  record.
```

```
library(ggplot2)
library(knitr)
r <- read.csv("data/module02/restaurant_ratings.csv")
A <- r$quality_rating == "Very Good"
B <- r$meal_price_usd >= 20 & r$meal_price_usd < 30
stopifnot(length(A) == 300, sum(B) == 118, sum(A) == 150, sum(A & B) == 64)
theme_set(theme_minimal(base_size = 13))
```

Conditional probability uses a restricted denominator

For $P(B) > 0$,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

The bar means **given**. It restricts the reference group to B. For our selection, 64 restaurants satisfy both conditions, while 118 satisfy the known price condition:

$$P(A \mid B) = \frac{64/300}{118/300} = \frac{64}{118} \approx 0.5424.$$

Reversing the condition changes the denominator:

$$P(B \mid A) = \frac{64}{150} \approx 0.4267.$$

```
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# rep(): repeats values; times repeats a vector and each repeats individual elements.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# nrow(): returns the number of rows in a table or matrix.
# The numerator counts the same intersection, but each denominator matches its condition.
# kable() reports both directions and the joint probability to make the distinction
 ↪  visible.
comparison <- data.frame(Question = c("A and B", "A given B", "B given A"),
                         Numerator = rep(sum(A & B), 3),
                         Denominator = c(nrow(r), sum(B), sum(A)))
comparison$Probability <- comparison$Numerator / comparison$Denominator
kable(comparison, digits = 4)
```

Question	Numerator	Denominator	Probability
A and B	64	300	0.2133
A given B	64	118	0.5424
B given A	64	150	0.4267

Interpretation: 54.24% of restaurants in that price band are Very Good; 42.67% of Very Good restaurants are in that band. Neither is the 21.33% share of all records that satisfy both conditions. If $P(B) = 0$, this elementary conditional probability is undefined; do not divide by zero.

Multiplication and independence

Rearranging the conditional formula gives the general multiplication rule:

$$P(A \cap B) = P(B)P(A \mid B).$$

Events are **independent** when $P(A \cap B) = P(A)P(B)$. If $P(B) > 0$, this is equivalent to $P(A \mid B) = P(A)$: knowing B leaves the probability of A unchanged.

```
# mean(): computes an average; logical TRUE/FALSE values contribute 1/0, giving a
 ↪  proportion.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# isTRUE(): returns TRUE only for a single nonmissing logical TRUE.
# Compare the actual joint probability with the product required by independence.
# all.equal() allows numerical tolerance; it does not perform a population hypothesis
 ↪  test.
joint <- mean(A & B)
product_if_independent <- mean(A) * mean(B)
kable(data.frame(Quantity = c("Actual P(A and B)", "P(A) times P(B)"),
                 Probability = c(joint, product_if_independent)), digits = 4)
```

Quantity	Probability
Actual P(A and B)	0.2133
P(A) times P(B)	0.1967

```
isTRUE(all.equal(joint, product_if_independent))
```

```
[1] FALSE
```

These events are dependent **for uniform selection from these 300 records**. That statement is an arithmetic property of the specified finite experiment, not an inferential test about a wider population.

> **i** Independent and mutually exclusive mean different things
>
> Mutually exclusive events cannot occur together. Independent events leave each other's probability unchanged. If two mutually exclusive events both have positive probability, they cannot be independent: their joint probability is zero, while the product of their probabilities is positive.

Selection without replacement

Suppose two records are selected successively and the first is not returned. If the first is Very Good, only **149 Very Good records among 299 remaining records** are available for the second selection. Thus,

$$P(\text{both Very Good}) = \frac{150}{300}\frac{149}{299}.$$

With replacement and independent selections, the corresponding probability is $(150/300)^2$. The mechanism determines whether the same probability can be multiplied twice.

Business case: which supplier produced a bad part?

In the book's manufacturing example, **65% of parts come from supplier 1 and 35% from supplier 2**. The stated bad-part probabilities are **2% for supplier 1 and 5% for supplier 2**. One purchased part is selected at random and found to be bad. What is the probability that it came from supplier 2? One outcome concerns one part; these are the textbook's probability inputs, not newly collected inspection observations.

في مثال التصنيع في الكتاب، يأتي **65% من القطع من المورّد الأول و35% من المورّد الثاني**. واحتمال أن تكون القطعة معيبة هو **2% لدى المورّد الأول و5% لدى المورّد الثاني**. تُختار قطعة مشتراة عشوائيًا ويتبيّن أنها معيبة. ما احتمال أن تكون من المورّد الثاني؟ تتعلق النتيجة بقطعة واحدة؛ وهذه احتمالات معطاة في الكتاب، وليست مشاهدات فحص جُمعت حديثًا.

Total probability and Bayes' theorem

Let S_1, S_2 denote the two suppliers and D a bad part. The suppliers are mutually exclusive and cover all parts. Multiply along each supplier-and-condition path, then add the disjoint bad-part paths:

$$P(D) = P(S_1)P(D \mid S_1) + P(S_2)P(D \mid S_2).$$

Bayes' theorem restricts those paths to the ones consistent with the new information:

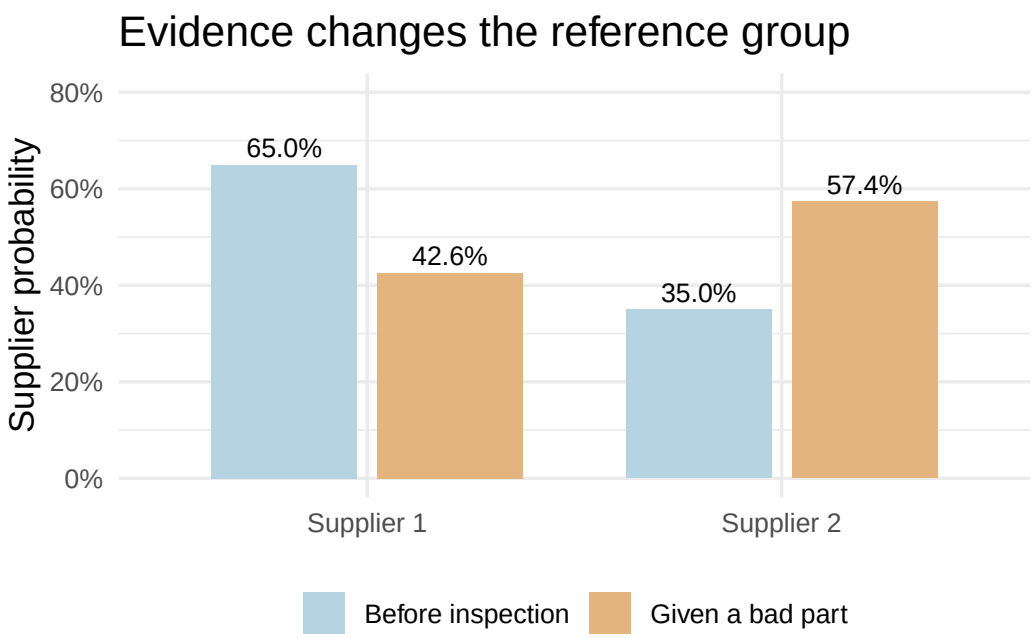
$$P(S_2 \mid D) = \frac{P(S_2)P(D \mid S_2)}{P(D)}.$$

```
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# These priors and bad-part rates are stated model inputs from textbook p.201, not raw
 ↪  data.
# Multiply within each supplier path, then divide by the total bad-part probability.
suppliers <- data.frame(Supplier = c("Supplier 1", "Supplier 2"),
                        Prior = c(0.65, 0.35), Bad_given_supplier = c(0.02, 0.05))
suppliers$Joint_bad <- suppliers$Prior * suppliers$Bad_given_supplier
p_bad <- sum(suppliers$Joint_bad)
suppliers$Supplier_given_bad <- suppliers$Joint_bad / p_bad
stopifnot(abs(p_bad - 0.0305) < 1e-12,
          abs(sum(suppliers$Supplier_given_bad) - 1) < 1e-12)
kable(suppliers, digits = 4)
```

Supplier	Prior	Bad_given_supplier	Joint_bad	Supplier_given_bad
Supplier 1	0.65	0.02	0.0130	0.4262
Supplier 2	0.35	0.05	0.0175	0.5738

The probability of a bad part is **0.0305**. Among bad parts, supplier 2 accounts for $0.0175/0.0305 \approx 0.5738$, or **57.38%**. This reverses $P(D \mid S_2) = 0.05$; the two conditional probabilities answer different questions. Supplier 2 has a smaller prior share but a higher stated bad-part rate.

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# rep(): repeats values; times repeats a vector and each repeats individual elements.
# c(): combines values into a vector, converting them to a common type when needed.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
# position_dodge(): places groups side by side; width controls their horizontal
 ↪  separation.
# geom_text(): draws labels from mapped data; position and hjust/vjust control placement.
# sprintf(): formats text using placeholders such as %.2f for two decimal places.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# paste0(): joins text with no separator; collapse can combine a vector into one string.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# theme(): customizes non-data plot elements such as legends, grids, and axis text.
# Compare supplier shares before inspection with shares conditional on a bad part.
# Fixed y limits and direct labels keep the prior/posterior comparison on the same scale.
plot_data <- data.frame(Supplier = rep(suppliers$Supplier, 2),
                        Information = rep(c("Before inspection", "Given a bad part"), each =
                          ↪  2),
                        Probability = c(suppliers$Prior, suppliers$Supplier_given_bad))
ggplot(plot_data, aes(Supplier, Probability, fill = Information)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7) +
  geom_text(aes(label = sprintf("%.1f%%", 100 * Probability)),
            position = position_dodge(width = 0.8), vjust = -0.4, size = 3.5) +
  scale_fill_manual(values = c("#B6D3E1", "#E4B47E")) +
  scale_y_continuous(limits = c(0, 0.8), labels = function(x) paste0(100 * x, "%")) +
  labs(x = NULL, y = "Supplier probability", fill = NULL,
       title = "Evidence changes the reference group") +
  theme(legend.position = "bottom")
```



Original R chart calculated from the book's two-supplier probability inputs, Table 4.6 and accompanying text, p.201.

A calculation sequence to remember

1. Identify the event and the information being conditioned on.
2. Write the relevant joint paths and multiply their probabilities.

3. Add disjoint paths to obtain the denominator.
4. Divide the relevant joint probability by that denominator.
5. Interpret the result in the restricted group and state the input assumptions.

Complete Lesson 2 practice, then continue to discrete random variables.

Sources and limits

Anderson et al. (2020), 8th ed., §4.4, pp.193–198, and §4.5, pp.201–205. Restaurant calculations reuse the checked CSV, with the documented aggregate verification limits. Supplier shares and conditional rates were checked directly against the local book PDF, p.201; no individual parts or inspection sample sizes are supplied or invented. Historical rates are accepted as inputs to the example, not established as current supplier performance.

Module 3: Conditional Probability — Practice

Lesson 2 | Calculate, justify, and interpret

Table of contents

Preparation 1
1. Which reference group? 1
2. Independence and exclusion 1
3. Two selections 1
4. Supplier evidence 2

Preparation

Read Lesson 2. Use its Restaurant events and supplier model inputs, retaining their stated business settings and assumptions. Refer to the English–Arabic introductions there; no additional observations are provided here.

1. Which reference group?

Using the Restaurant CSV, calculate $P(A \mid B)$, $P(B \mid A)$, and $P(A \cap B)$. Show the numerator and denominator for each. Interpret each in one sentence.

2. Independence and exclusion

1. Compare $P(A \cap B)$ with $P(A)P(B)$. Are the events independent for uniform selection from these records?
2. Explain why this is not an independence hypothesis test for a larger population.
3. Explain why Very Good and Excellent ratings are mutually exclusive but are not independent when both have positive probability.

3. Two selections

Find the probability that two records are both Very Good when sampling without replacement. Compare it with independent selection with replacement. State the changed second-selection denominator and numerator.

4. Supplier evidence

Use supplier shares 0.65 and 0.35 and bad-part rates 0.02 and 0.05 from the lesson.

1. Calculate each supplier-and-bad joint probability and the total probability of a bad part.
2. Calculate both supplier probabilities conditional on a bad part and check that they sum to 1.
3. Calculate the probability a part comes from supplier 2 **given that it is good**.
4. Explain why 0.05 is not the answer to the question “Which supplier produced this bad part?”
5. State what raw inspection information is unavailable.

Submit a table of results with reference groups and assumptions. Return to the lesson.

Module 3: Discrete Random Variables

Lesson 3 | Turn outcomes into a probability distribution

Table of contents

What we will learn 1
Business case: daily sales at DiCarlo Motors 1
Discrete values and a probability mass function 2
Events and the cumulative distribution 3
Expected value and variability 4
Summary: a distribution is more than its mean 5
Sources and limits 5

What we will learn

A **random variable** assigns a numerical value to each outcome of an experiment. Uppercase X denotes the variable before the outcome is known; lowercase x denotes a possible value.

By the end, you should be able to construct a discrete probability mass function, calculate event and cumulative probabilities, and interpret expected value and variance.

Reading: main textbook, §§5.1–5.3, printed pp.218–228; DiCarlo Motors, pp.222–223 and Tables 5.4–5.6, pp.223, 227–228. Bivariate portfolio distributions in §5.4 are outside this module's selected sequence.

Business case: daily sales at DiCarlo Motors

The book describes **300 operating days at DiCarlo Motors in Saratoga, New York**. The numbers of days with **0, 1, 2, 3, 4, and 5 cars sold** were **54, 117, 72, 42, 12, and 3**, respectively. Let X be the number of cars sold on a day. How likely is a day with at least three sales, and what sales level should be expected on average? One observational unit is one operating day. Only the published frequency summary is available here; the original sequence of daily observations is not supplied.

يصف الكتاب **ثلاثمئة يوم عمل في معرض DiCarlo Motors في ساراتوغا بولاية نيويورك**. كانت أعداد الأيام التي بيعت فيها **0** و**1** و**2** و**3** و**4** و**5 سيارات** هي **54** و**117** و**72** و**42** و**12** و**3** على الترتيب. ليكن المتغير X عدد السيارات المبيعة في يوم واحد. ما احتمال بيع ثلاث سيارات على الأقل في اليوم، وما متوسط المبيعات المتوقع؟ الوحدة المشاهدة هي يوم عمل واحد. المتاح هنا هو ملخّص التكرارات المنشور فقط، ولا يتوافر الترتيب الأصلي للمشاهدات اليومية.

```
# library(): loads a package so its functions can be used in this session.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# all(): returns TRUE only when every supplied logical value is TRUE.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
```

```
# theme_set(): sets the default theme for subsequent ggplot charts in this R session.
# theme_minimal(): uses a minimal chart style; base_size controls the base text size in
 ↪  points.
# Enter the six published frequency cells from textbook p.222; do not expand them into
 ↪  invented days.
# Dividing by 300 creates the probability mass function verified against Table 5.4, p.223.
library(ggplot2)
library(knitr)
sales <- data.frame(Cars = 0:5, Days = c(54, 117, 72, 42, 12, 3))
sales$Probability <- sales$Days / sum(sales$Days)
stopifnot(sum(sales$Days) == 300,
          all(abs(sales$Probability - c(.18, .39, .24, .14, .04, .01)) < 1e-12))
theme_set(theme_minimal(base_size = 13))
```

Discrete values and a probability mass function

X is **discrete**: its possible values form a finite or countable set. Sales of 1.5 cars cannot occur on a single day. Flight duration is **continuous** in the model because it can vary over an interval, even if a measurement is rounded to minutes.

The probability mass function is

$$p(x) = P(X = x), \qquad p(x) \geq 0, \qquad \sum_{x} p(x) = 1.$$

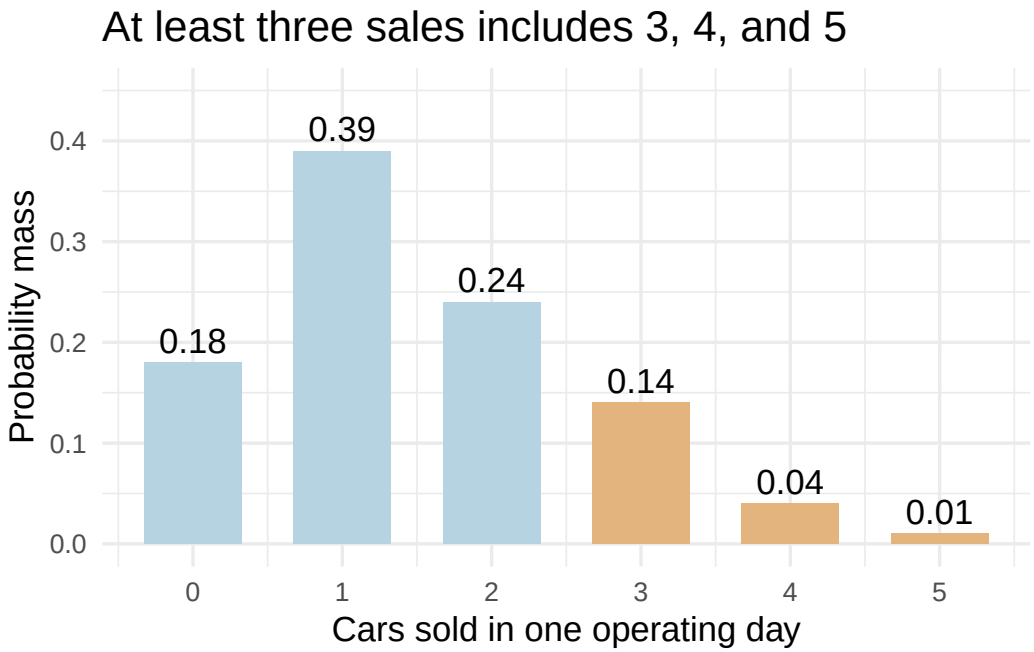
```
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# all(): returns TRUE only when every supplied logical value is TRUE.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# Each row is a published sales-count category, not an individual day or a newly created
 ↪  record.
# The probability column sums to one; kable() prints the six-cell summary.
stopifnot(all(sales$Probability >= 0), abs(sum(sales$Probability) - 1) < 1e-12)
kable(sales, digits = 2)
```

Cars	Days	Probability
0	54	0.18
1	117	0.39
2	72	0.24
3	42	0.14
4	12	0.04
5	3	0.01

For uniform selection of one of the 300 recorded days, these relative frequencies give exact selection probabilities. Using them for a **future day** is an empirical probability model that assumes those historical days remain relevant. It does not establish that more than five cars can never be sold in future operations.

```
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
# geom_text(): draws labels from mapped data; position and hjust/vjust control placement.
# sprintf(): formats text using placeholders such as %.2f for two decimal places.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# c(): combines values into a vector, converting them to a common type when needed.
```

```r
# scale_x_continuous(): controls numeric x-axis breaks, labels, limits, and spacing.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# Highlight the possible values in the event X >= 3; the event probability is their summed
 ↪  mass.
# Bars are separated because X takes integer counts; labels give the probability at each
 ↪  count.
ggplot(sales, aes(Cars, Probability, fill = Cars >= 3)) +
  geom_col(width = 0.65) +
  geom_text(aes(label = sprintf("%.2f", Probability)), vjust = -0.4) +
  scale_fill_manual(values = c("#B6D3E1", "#E4B47E"), guide = "none") +
  scale_x_continuous(breaks = 0:5) +
  scale_y_continuous(limits = c(0, 0.45)) +
  labs(x = "Cars sold in one operating day", y = "Probability mass",
       title = "At least three sales includes 3, 4, and 5")
```



Original R chart of the six probabilities in textbook Table 5.4. Orange bars identify the event; they do not represent additional observations.

Events and the cumulative distribution

Sum masses at all values in the event:

$$P(X \geq 3) = p(3) + p(4) + p(5) = 0.19.$$

The **cumulative distribution function**, or CDF, is $F(t) = P(X \leq t)$. It accumulates probabilities up to and including its argument. For an integer-valued X,

$$P(X > k) = 1 - F(k), \qquad P(X \geq k) = 1 - F(k-1).$$

```r
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# c(): combines values into a vector, converting them to a common type when needed.
# sum(): adds vector entries; on logical values it counts TRUE entries.
# cumsum() accumulates mass in increasing Cars order to obtain F(x) = P(X <= x).
# Indexing by a condition selects all mass values in an event before summing them.
```

```r
sales$CDF <- cumsum(sales$Probability)
kable(sales[c("Cars", "Probability", "CDF")], digits = 2)
```

Cars	Probability	CDF
0	0.18	0.18
1	0.39	0.57
2	0.24	0.81
3	0.14	0.95
4	0.04	0.99
5	0.01	1.00

```r
c("Exactly three" = sum(sales$Probability[sales$Cars == 3]),
  "At least three" = sum(sales$Probability[sales$Cars >= 3]),
  "More than three" = sum(sales$Probability[sales$Cars > 3]))
```

```
  Exactly three  At least three More than three
           0.14            0.19            0.05
```

“Exactly three” has probability 0.14; “at least three” has probability 0.19; “more than three” has probability 0.05. Draw or list the included values before choosing a formula.

Expected value and variability

The **expected value** is a probability-weighted mean:

$$\mu = E(X) = \sum_x xp(x).$$

For this distribution, $E(X) = 1.50$ cars per day. This is an average over repetitions under the model, not a possible single-day sales count, a guaranteed forecast, or the most probable value. The **mode** is 1 car, whose probability is 0.39.

The variance and standard deviation summarize dispersion around the model mean:

$$\mathrm{Var}(X) = \sum_x (x-\mu)^2 p(x), \qquad \sigma = \sqrt{\mathrm{Var}(X)}.$$

```r
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# sum(x*p) and sum((x-mu)^2*p) use probability weights, not the sample n-1 correction.
# sqrt() returns SD in cars per day; variance is in squared cars for the daily count.
mu <- sum(sales$Cars * sales$Probability)
variance <- sum((sales$Cars - mu)^2 * sales$Probability)
kable(data.frame(Measure = c("Expected cars per day", "Variance", "SD"),
                 Value = c(mu, variance, sqrt(variance))), digits = 4)
```

Measure	Value
Expected cars per day	1.500
Variance	1.250
SD	1.118

```
stopifnot(abs(mu - 1.5) < 1e-12, abs(variance - 1.25) < 1e-12)
```

Do not apply `var(sales$Cars)` to obtain this model variance: that would weight the six possible values equally and apply a sample correction. It ignores the given probabilities.

For 30 operating days with the same daily expected value, expected total sales are $30(1.5) = 45$ cars. **Linearity of expectation does not require independence.** Calculating the variance of the total as 30 times the daily variance would additionally require zero covariance between daily counts; independence would be sufficient.

Summary: a distribution is more than its mean

Question	Use	DiCarlo answer
What values are possible in the stated model?	Support	0–5 cars
What is most probable?	Mode	1 car
How likely are at least three sales?	Sum the relevant masses	0.19
What average does the model imply?	Expected value	1.50 cars/day
How variable is the daily count?	Model SD	About 1.118 cars

Complete Lesson 3 practice, then continue to binomial and Poisson models.

Sources and limits

Anderson et al. (2020), 8th ed., §§5.1–5.3. The six aggregate counts were transcribed from printed p.222 and checked against all six probabilities in Table 5.4, p.223; mean 1.50 and variance 1.25 match Tables 5.5–5.6, pp.227–228. These are published aggregate inputs, not a converted workbook or a 300-row daily dataset. No day ordering, dates, current sales conditions, or independent-day assumption is established by the summary.

Module 3: Discrete Random Variables — Practice

Lesson 3 | Calculate, justify, and interpret

Table of contents

Preparation 1
1. Build the distribution 1
2. Translate events 1
3. Use a cumulative table 1
4. Center and spread 2
5. Extend with care 2

Preparation

Read Lesson 3, including the DiCarlo English–Arabic introduction and the distinction between the published aggregate table and individual observations.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# Enter the six published count categories, not a fabricated sequence of individual sales
 ↪  days.
# Each Days value is a frequency; divide by their total to obtain probability weights.
sales <- data.frame(Cars = 0:5, Days = c(54, 117, 72, 42, 12, 3))
```

1. Build the distribution

Define X and its support. Calculate the probability mass function and verify nonnegative masses summing to 1. Is 1.5 a possible realization of X?

2. Translate events

Calculate the probabilities of exactly two cars, at most two cars, at least two cars, more than two cars, and between one and three cars inclusive. List the included count values before computing each answer.

3. Use a cumulative table

Add a CDF column. Find the smallest sales count whose cumulative probability is at least 0.50. Explain how this differs from calculating the expected value.

4. Center and spread

Find the model mean, variance, SD, and mode. Explain why applying `mean()` or `var()` directly to the six values `0:5` ignores important information. State which quantities have cars, squared cars, or no units.

5. Extend with care

Find expected total sales for 30 operating days if the daily expectation remains the same. Does that expected-total calculation require independence? What extra condition would justify multiplying the daily variance by 30? Explain why the summary does not establish a five-car upper limit for future operations.

Submit a table, a probability plot, and your interpretations. Return to the lesson.

Module 3: Binomial and Poisson Models

Lesson 4 | Select a count model and calculate the right event

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Business case: Martin Clothing Store purchases . . . 1
Check the binomial assumptions . . . 2
Simulation checks an implication of the model . . . 4
Business case: arrivals in an interval . . . 4
The Poisson model uses an interval mean . . . 5
Choose by the mechanism . . . 6
Sources and limits . . . 7

What we will learn

A named probability distribution connects a set of assumptions to a probability mass function. We use a **binomial model** for successes in a fixed number of trials and a **Poisson model** for occurrences within a fixed exposure interval.

By the end, you should be able to justify the model, translate a verbal event into an inequality, calculate its probability in R, and interpret simulation as a model illustration.

Reading: main textbook, §5.5, printed pp.242–249 (Martin Clothing Store, p.244), and §5.6, pp.252–255 (arrivals example, p.253). The hypergeometric distribution in §5.7 is a later extension for sampling without replacement; do not assume a binomial model automatically applies there.

```
# library(): loads a package so its functions can be used in this session.
# theme_set(): sets the default theme for subsequent ggplot charts in this R session.
# theme_minimal(): uses a minimal chart style; base_size controls the base text size in
 ↪  points.
# Load display tools and choose a common chart theme; calculations use base R
 ↪  distributions.
# No observation file is imported: the following examples use stated textbook model
 ↪  parameters.
library(ggplot2)
library(knitr)
theme_set(theme_minimal(base_size = 13))
```

Business case: Martin Clothing Store purchases

At **Martin Clothing Store**, the book asks about the next **three customers**. The stated probability that a customer purchases is **0.30**. Assume that probability is the same for each customer and that the three purchase decisions are independent. What is the probability that exactly two customers purchase? Each trial concerns one customer; the random variable counts purchases among the three customers.

في متجر **Martin Clothing Store**، يتناول سؤال الكتاب **الزبائن الثلاثة القادمين**. احتمال أن يشتري الزبون هو **0.30** وفق المعطيات. نفترض ثبات هذا الاحتمال لكل زبون واستقلال قرارات الشراء الثلاثة. ما احتمال أن يشتري زبونان بالضبط؟ تتعلق كل محاولة بزبون واحد، ويعدّ المتغير العشوائي المشترين بين الزبائن الثلاثة.

Check the binomial assumptions

Let X count purchases. A binomial model requires:

1. A fixed number of trials, $n = 3$.
2. Two outcomes on each trial: purchase or no purchase.
3. The same success probability on every trial, $p = 0.30$.
4. Independent trial outcomes.

“Success” is a mathematical label for the outcome being counted. It need not mean a desirable outcome in other applications. A shopping group making a joint purchase decision, or a promotion that changes during observation, could violate these assumptions.

We write $X \sim \text{Binomial}(n, p)$, with

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \dots, n.$$

Exactly two purchases can occur in three arrangements: purchase–purchase–no purchase, purchase–no purchase–purchase, or no purchase–purchase–purchase. Each arrangement has probability $0.3^2(0.7)$, so the total is $3(0.3^2)(0.7) = 0.189$.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# dbinom(x, size, prob) returns P(X=x); choose(3,2) counts arrangements of two successes.
# size is the fixed number of customers, while prob is the probability for one customer.
binomial <- data.frame(Purchases = 0:3, Probability = dbinom(0:3, size = 3, prob = 0.30))
kable(binomial, digits = 3)
```

Purchases	Probability
0	0.343
1	0.441
2	0.189
3	0.027

```
stopifnot(abs(dbinom(2, 3, .30) - choose(3, 2) * .30^2 * .70) < 1e-12)
```

Exact, cumulative, and upper-tail probabilities

```
# c(): combines values into a vector, converting them to a common type when needed.
# dbinom(): returns binomial point probabilities for size trials and success probability
 ↪  prob.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
```

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# names(): reads or assigns names to vector elements or table columns.
# unname(): removes names or dimension names without changing the underlying values.
# pbinom(k,...) gives P(X<=k); lower.tail=FALSE instead gives P(X>k).
# At least two means X>1, so its upper-tail call uses 1 rather than 2.
answers <- c("Exactly two" = dbinom(2, size = 3, prob = .30),
             "At most two" = pbinom(2, size = 3, prob = .30),
             "At least two" = pbinom(1, size = 3, prob = .30, lower.tail = FALSE),
             "At least one" = pbinom(0, size = 3, prob = .30, lower.tail = FALSE))
kable(data.frame(Event = names(answers), Probability = unname(answers)), digits = 3)
```

Event	Probability
Exactly two	0.189
At most two	0.973
At least two	0.216
At least one	0.657

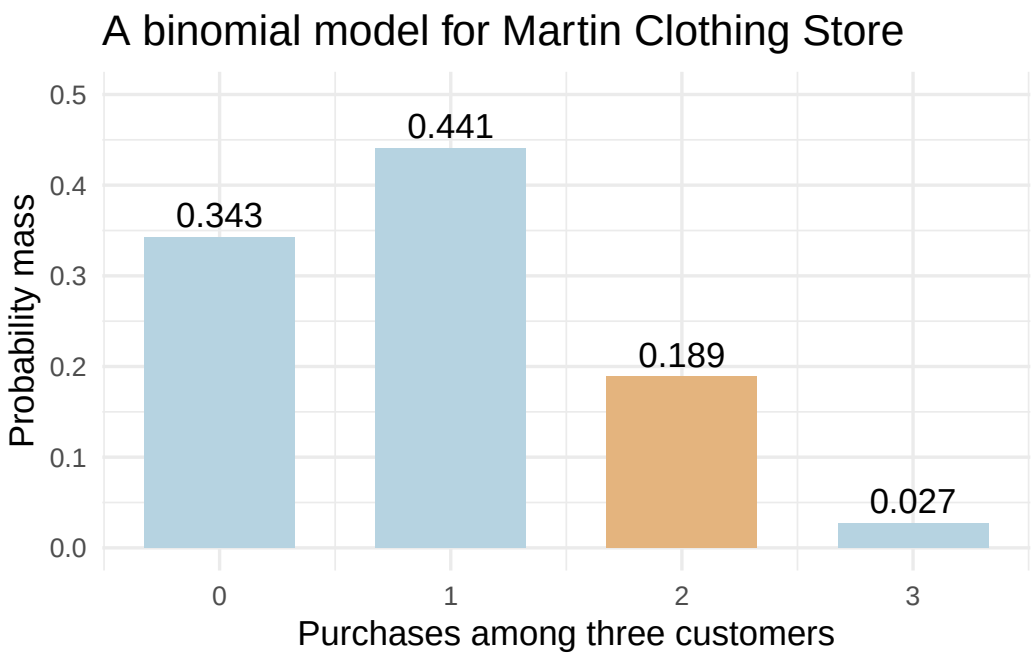
The probability of exactly two purchases is **0.189**; at least two has probability **0.216**. Using `pbinom(2, ..., lower.tail=FALSE)` would calculate more than two, which is a different event.

For a binomial count,

$$E(X) = np = 0.9, \qquad \text{Var}(X) = np(1-p) = 0.63.$$

The SD is $\sqrt{0.63} \approx 0.794$ purchases. The expected count 0.9 is an average across repetitions, not the purchase probability for a single customer and not a possible count for one three-customer group.

```r
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
# geom_text(): draws labels from mapped data; position and hjust/vjust control placement.
# sprintf(): formats text using placeholders such as %.2f for two decimal places.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# c(): combines values into a vector, converting them to a common type when needed.
# scale_x_continuous(): controls numeric x-axis breaks, labels, limits, and spacing.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# The four bars exhaust this binomial distribution; their masses sum to one.
# Highlighting exactly two matches the business question rather than implying observed
 ↪  frequencies.
ggplot(binomial, aes(Purchases, Probability, fill = Purchases == 2)) +
  geom_col(width = .65) +
  geom_text(aes(label = sprintf("%.3f", Probability)), vjust = -.4) +
  scale_fill_manual(values = c("#B6D3E1", "#E4B47E"), guide = "none") +
  scale_x_continuous(breaks = 0:3) +
  scale_y_continuous(limits = c(0, .50)) +
  labs(x = "Purchases among three customers", y = "Probability mass",
       title = "A binomial model for Martin Clothing Store")
```



Original R model plot using the book's $n = 3$ and $p = 0.30$. These bars are probabilities, not observed customer counts.

Simulation checks an implication of the model

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# dbinom(): returns binomial point probabilities for size trials and success probability
 ↪  prob.
# set.seed() makes the simulated illustration repeatable; rbinom() generates counts under
 ↪  the model.
# mean(simulated == 2) estimates the event probability, not the real store's unknown
 ↪  purchase rate.
set.seed(307312)
simulated <- rbinom(10000, size = 3, prob = .30)
kable(data.frame(Method = c("Exact model", "10,000 simulated groups"),
                 Probability_of_two = c(dbinom(2, 3, .30), mean(simulated == 2))), digits
                  ↪  = 4)
```

Method	Probability_of_two
Exact model	0.1890
10,000 simulated groups	0.1907

The simulated proportion should be near 0.189 but need not equal it. More repetitions usually reduce simulation variability. Agreement with a simulation generated from the same assumptions **does not validate those assumptions for real customers**.

Business case: arrivals in an interval

The book considers arrivals at a **large hospital emergency room on weekday mornings**, with an average of **10 patients per 15-minute interval**. Treat the example as a count-model

exercise: assume a constant arrival rate, independent counts in disjoint intervals, and individual arrivals rather than batches. What is the probability of exactly five arrivals in 15 minutes, or exactly one arrival in three minutes? The counted event is one patient arrival; the exposure is a specified time interval. These stated inputs do not establish a current operational forecast.

يتناول الكتاب وصول المرضى إلى **قسم الطوارئ في مستشفى كبير صباح أيام العمل**، بمتوسط **عشرة مرضى خلال كل فترة مدتها خمس عشرة دقيقة**. نستخدم المثال للتدرّب على نموذج العدّ: نفترض ثبات معدل الوصول واستقلال الأعداد في الفترات غير المتداخلة، ووصول الأفراد دون دفعات جماعية. ما احتمال وصول خمسة مرضى بالضبط خلال خمس عشرة دقيقة، أو مريض واحد بالضبط خلال ثلاث دقائق؟ الحدث المعدود هو وصول مريض واحد، ومدة التعرّض هي الفترة الزمنية المحددة. لا تثبت هذه المعطيات توقعًا تشغيليًا حاليًا.

The Poisson model uses an interval mean

For a homogeneous Poisson process, a constant rate, independent increments, and negligible probability of multiple arrivals in a sufficiently short interval imply Poisson counts. Merely having a count variable or knowing its mean is insufficient.

If $X \sim \text{Poisson}(\lambda)$,

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \dots,$$

and $E(X) = \text{Var}(X) = \lambda$. Here, λ is the **mean count for the specified interval**. The book writes this mean as μ; R calls the argument `lambda`.

For 15 minutes, $\lambda = 10$. For three minutes at the same rate,

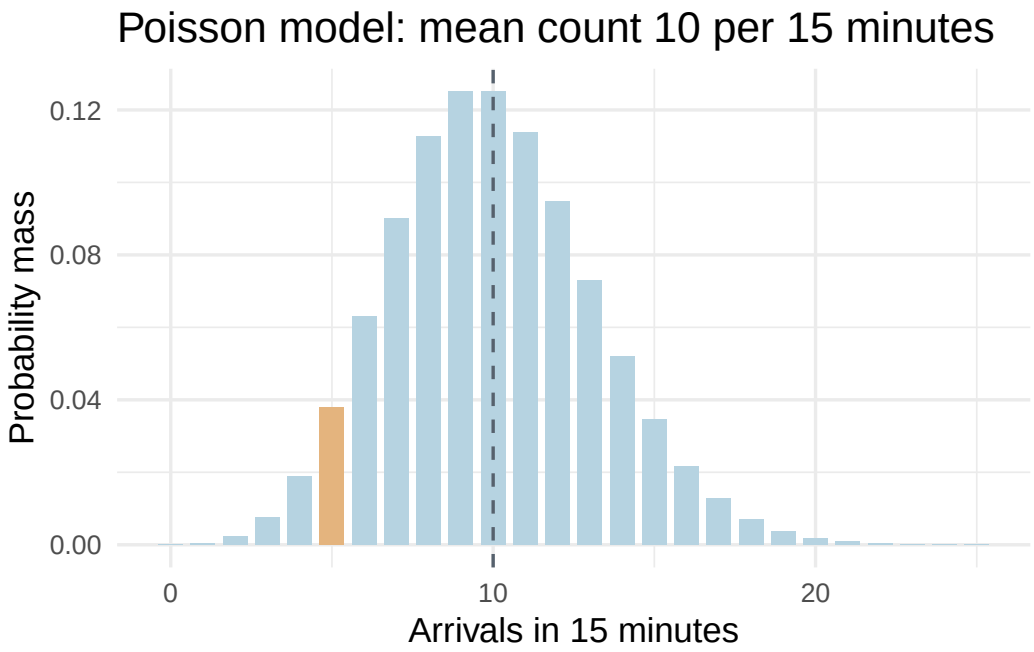
$$\lambda_{3\text{ min}} = 10 \times \frac{3}{15} = 2.$$

```
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# names(): reads or assigns names to vector elements or table columns.
# unname(): removes names or dimension names without changing the underlying values.
# dpois() returns an exact-count mass; ppois(k, lower.tail=FALSE) returns P(X>k).
# Scale lambda with the interval length before calculating a three-minute probability.
lambda_15 <- 10
lambda_3 <- lambda_15 * 3 / 15
poisson_answers <- c("Exactly 5 in 15 minutes" = dpois(5, lambda = lambda_15),
                     "More than 10 in 15 minutes" = ppois(10, lambda = lambda_15,
                      ↪  lower.tail = FALSE),
                     "Exactly 1 in 3 minutes" = dpois(1, lambda = lambda_3))
kable(data.frame(Event = names(poisson_answers), Probability = unname(poisson_answers)),
 ↪  digits = 4)
```

Event	Probability
Exactly 5 in 15 minutes	0.0378
More than 10 in 15 minutes	0.4170
Exactly 1 in 3 minutes	0.2707

The exact-five probability is about **0.0378**. The exact-one probability over three minutes is about **0.2707**. Keeping `lambda=10` for three minutes would use the wrong exposure.

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# dpois(): returns Poisson count probabilities; lambda is the mean for the stated
 ↪  exposure.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
# geom_vline(): adds vertical references at xintercept; linetype selects solid or dashed
 ↪  lines.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# c(): combines values into a vector, converting them to a common type when needed.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# Plot a finite window of an infinite-support distribution; the omitted upper tail is not
 ↪  zero.
# The dashed line marks the model mean, while the highlighted bar marks exactly five
 ↪  arrivals.
poisson <- data.frame(Arrivals = 0:25, Probability = dpois(0:25, lambda = 10))
ggplot(poisson, aes(Arrivals, Probability, fill = Arrivals == 5)) +
  geom_col(width = .75) +
  geom_vline(xintercept = 10, linetype = "dashed", color = "#56616D") +
  scale_fill_manual(values = c("#B6D3E1", "#E4B47E"), guide = "none") +
  labs(x = "Arrivals in 15 minutes", y = "Probability mass",
       title = "Poisson model: mean count 10 per 15 minutes")
```



Original R model plot. Only counts 0–25 are displayed; the model also allows larger counts. The dashed line is the mean, not a capacity limit.

Choose by the mechanism

Question	Binomial	Poisson
What is fixed?	Number of trials	Exposure interval
What is counted?	Successes among trials	Occurrences during exposure
Parameters	n, p	Mean count λ for that exposure
Possible counts	0 through n	Any nonnegative integer

Question	Binomial	Poisson
Main assumptions	Independent trials, common p	Constant-rate process, independent increments, no batches

A small count by itself does not select a model. Discuss the process and assumptions first. Complete Lesson 4 practice, then continue to continuous distributions.

Sources and limits

Anderson et al. (2020), 8th ed., §5.5, especially p.244, and §5.6, pp.252–255. Parameters were checked directly against the local PDF. No customer records or arrival timestamps are available here. Simulation generates explicitly artificial groups under the binomial model; it does not replace missing publisher data or establish model fit.

Module 3: Binomial and Poisson Models — Practice

Lesson 4 | Calculate, justify, and interpret

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2. Translate binomial events 1
3. Change the exposure 1
4. Simulate and interpret 2
5. Sampling without replacement 2

Preparation

Read Lesson 4. Use the Martin Clothing Store and arrivals inputs already introduced in English and Arabic. These are model calculations rather than an analysis of newly supplied observations.

1. Select the model

Explain why the store example can be binomial with $n = 3$ and $p = 0.30$, and why the arrival count can be Poisson with mean 10 per 15 minutes. Give one plausible way each model's assumptions could fail.

2. Translate binomial events

Calculate the probability of exactly two, at most two, at least two, and more than two purchases. Then find the expected count, variance, and SD. Explain the difference between `dbinom()` and `pbinom()` and the boundary needed for "at least two."

3. Change the exposure

For the same stated arrival rate, find:

1. The mean count in three minutes.
2. The probability of no arrivals in three minutes.
3. The probability of at least two arrivals in three minutes.
4. The probability of more than ten arrivals in fifteen minutes.

Explain why the model mean must be tied to the interval and why the count has no fixed upper bound in the Poisson model.

4. Simulate and interpret

Use a fixed seed and simulate 10,000 three-customer binomial counts. Estimate the probability of at least two purchases and compare it with the exact value. Explain why a different seed may change the estimate and why this comparison does not establish model fit for real customers.

5. Sampling without replacement

Explain why taking three records from a finite list without replacement need not give independent trials with the same success probability. Do not apply a binomial model merely because there are three selections and two possible labels.

Submit a method-and-event table with results and a short interpretation. Return to the lesson.

Module 3: Continuous Distributions

Lesson 5 | Read areas, tails, and quantiles

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Calculate a tail or an interval . . . 4
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What we will learn

A continuous model assigns probabilities to **intervals through area under a density curve**. We use the uniform distribution to make area concrete, then the normal distribution to calculate tails and choose a cutoff.

By the end, you should be able to distinguish density from probability, standardize a normal value, calculate interval and tail probabilities, and invert a cumulative probability to find a quantile.

Reading: main textbook, §6.1, printed pp.275–277, and §6.2, pp.279–290. The exponential distribution (§6.3, pp.293–298) is optional follow-up reading and is not developed in this selected sequence.

```
# library(): loads a package so its functions can be used in this session.
# theme_set(): sets the default theme for subsequent ggplot charts in this R session.
# theme_minimal(): uses a minimal chart style; base_size controls the base text size in
 ↪  points.
# Load chart/table tools; the examples below use parameters stated in the textbook.
# seq() creates plotting grids only; these grids are not observed flight or tire records.
library(ggplot2)
library(knitr)
theme_set(theme_minimal(base_size = 13))
```

Business case: a flight-time interval

The textbook models a flight from **Chicago to New York** with duration between **120 and 140 minutes**, assigning equal probability to intervals of equal length within those bounds. If T is the duration of one flight in minutes, what is the probability that $128 \leq T \leq 136$? This is the book's uniform model; individual flight measurements are not supplied here.

يستخدم الكتاب نموذجًا لمدة رحلة جوية من **شيكاغو إلى نيويورك**، بحيث تقع المدة بين **120 و140 دقيقة**، وتتساوى احتمالات الفترات المتساوية في الطول داخل هذين الحدّين. إذا كان T مدة رحلة واحدة بالدقائق، فما احتمال أن تكون المدة بين 128 و136 دقيقة؟ هذا هو النموذج المنتظم المعطى في الكتاب؛ ولا تتوافر هنا قياسات الرحلات الفردية.

Probability is area, not height

For a continuous variable with density $f(t)$:

- The density is nonnegative and its total area is 1.
- $P(a \le T \le b)$ is the area under the density between a and b.
- $P(T = t) = 0$ at any single exact point in a continuous model.
- Consequently, including or excluding an endpoint does not change a continuous interval probability.

The density height can exceed 1 if the distribution is concentrated in a narrow interval. It has units inverse to the variable, such as **per minute**; the resulting area is a unit-free probability.

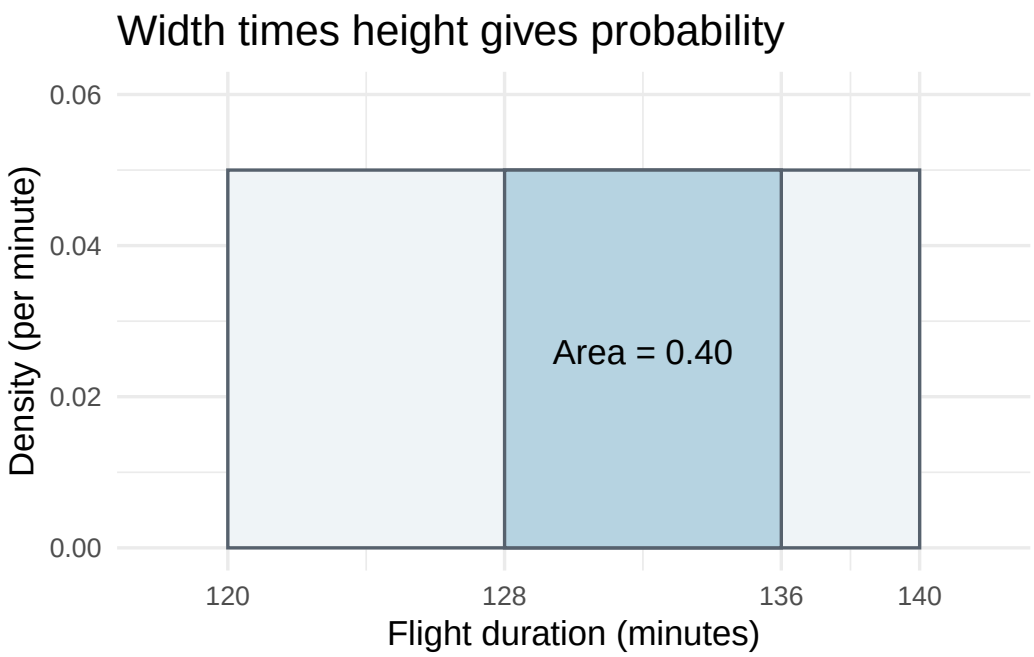
For $T \sim \text{Uniform}(120, 140)$, the density is $1/(140 - 120) = 0.05$ per minute inside the bounds and zero outside them. Therefore,

$$P(128 \le T \le 136) = (136 - 128) \times 0.05 = 0.40.$$

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# punif(t,min,max) gives P(T<=t); subtract cumulative areas for the requested interval.
# dunif() returns density height, whereas an exact point has zero continuous probability.
flight_probability <- punif(136, min = 120, max = 140) - punif(128, min = 120, max = 140)
kable(data.frame(Quantity = c("Density at 130 (per minute)", "Probability from 128 to
 ↪  136", "Probability at exactly 130"),
                 Value = c(dunif(130, 120, 140), flight_probability, 0)), digits = 4)
```

Quantity	Value
Density at 130 (per minute)	0.05
Probability from 128 to 136	0.40
Probability at exactly 130	0.00

```
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# annotate(): adds a fixed plot annotation rather than mapping one label per data row.
# scale_x_continuous(): controls numeric x-axis breaks, labels, limits, and spacing.
# c(): combines values into a vector, converting them to a common type when needed.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# The rectangle height is 1/20 per minute; its highlighted width is eight minutes.
# geom_rect() draws model areas, not histogram bars or recorded flights.
ggplot() +
  geom_rect(aes(xmin = 120, xmax = 140, ymin = 0, ymax = .05), fill = "#EFF4F7", color =
   ↪  "#56616D") +
  geom_rect(aes(xmin = 128, xmax = 136, ymin = 0, ymax = .05), fill = "#B6D3E1", color =
   ↪  "#56616D") +
  annotate("text", x = 132, y = .026, label = "Area = 0.40") +
  scale_x_continuous(breaks = c(120, 128, 136, 140), limits = c(118, 142)) +
  scale_y_continuous(limits = c(0, .06)) +
  labs(x = "Flight duration (minutes)", y = "Density (per minute)",
       title = "Width times height gives probability")
```



Original R illustration calculated from the uniform flight-time model, textbook pp.275–277.

A recorded duration of “130 minutes” may represent a **rounded interval**, not the exact mathematical point 130. For example, nearest-minute rounding could correspond to $129.5 \leq T < 130.5$, which has positive probability. Recording precision and the underlying continuous variable are different concepts.

The normal model and standardization

A normal distribution is symmetric and bell-shaped. Its parameters are the mean μ and SD $\sigma > 0$. We write $X \sim N(\mu, \sigma^2)$; the second quantity in this notation is **variance**, but R’s sd argument requires **standard deviation**.

Standardization expresses a value in SD units:

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim N(0, 1).$$

This connects to Module 2’s z-scores. A standardized score alone does not make an arbitrary variable normally distributed. Normal tail calculations require a normal model to be reasonable.

Business case: Grear tire lifetimes

Grear Tire Company considers a mileage guarantee for a new steel-belted radial tire. The textbook reports an estimated mean lifetime of **36,500 miles** and SD **5,000 miles**, and treats a normal distribution as reasonable. What proportion lasts more than **40,000 miles**? What mileage cutoff places **10% of lifetimes below it**? The variable is lifetime in miles for one tire. The raw road-test observations are not available here, so we use the stated model rather than independently checking its fit.

تدرس شركة **Grear Tire Company** ضمانًا لعدد الأميال لإطار شعاعي جديد مدعّم بالفولاذ. يذكر الكتاب متوسط عمر مقدّرًا قدره **36,500 ميل** وانحرافًا معياريًا قدره **5,000 ميل**، ويعدّ التوزيع الطبيعي نموذجًا معقولًا. ما نسبة الإطارات التي يتجاوز عمرها **40,000 ميل**؟ وما حدّ الأميال الذي يقع **10% من الأعمار دونه**؟ المتغير هو عمر إطار واحد بالأميال. لا تتوافر هنا مشاهدات اختبارات الطريق الأصلية، لذا نستخدم النموذج المعطى دون التحقق من ملاءمته بصورة مستقلة.

Calculate a tail or an interval

For 40,000 miles,

$$z = \frac{40{,}000 - 36{,}500}{5{,}000} = 0.70.$$

```
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# names(): reads or assigns names to vector elements or table columns.
# unname(): removes names or dimension names without changing the underlying values.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# pnorm() returns a lower-tail area; lower.tail=FALSE directly returns the upper-tail
 ↪  area.
# sd=5000 supplies SD in miles, not variance; subtract two CDF values for an interval.
mu <- 36500
sigma <- 5000
z <- (40000 - mu) / sigma
normal_answers <- c("More than 40,000 miles" = pnorm(40000, mean = mu, sd = sigma,
 ↪  lower.tail = FALSE),
                    "20,000 to 40,000 miles" = pnorm(40000, mu, sigma) - pnorm(20000, mu,
                     ↪  sigma))
kable(data.frame(Event = names(normal_answers), Probability = unname(normal_answers)),
 ↪  digits = 4)
```

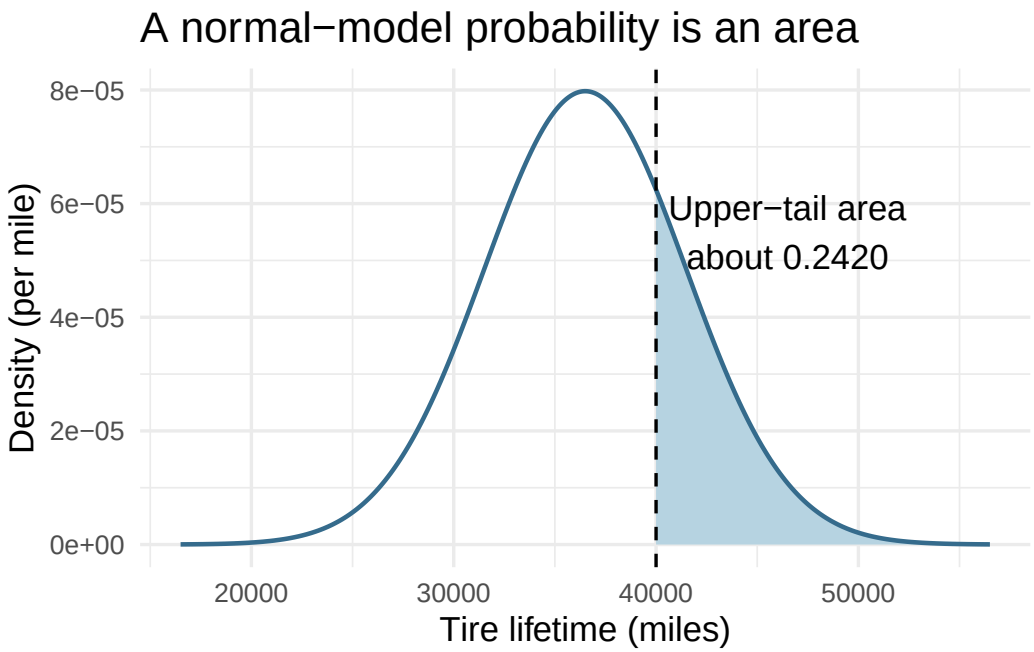
Event	Probability
More than 40,000 miles	0.2420
20,000 to 40,000 miles	0.7576

```
stopifnot(abs(pnorm(z, lower.tail = FALSE) - normal_answers[1]) < 1e-12)
```

About **24.20%** exceed 40,000 miles under the stated model. The probability is an area; `dnorm(40000, ...)` would give a density height, not this tail probability.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# sort(): sorts values; decreasing=TRUE reverses the default ascending order.
# unique(): keeps distinct values or rows in first-occurrence order.
# c(): combines values into a vector, converting them to a common type when needed.
# seq(): creates an equally spaced sequence; by sets spacing and length.out sets the
 ↪  number of values.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_area(): fills the area between a curve and zero.
# subset(): keeps entries or rows satisfying a condition; select can choose columns.
# geom_line(): joins observations in x order; group determines which observations form
 ↪  each line.
# geom_vline(): adds vertical references at xintercept; linetype selects solid or dashed
 ↪  lines.
# annotate(): adds a fixed plot annotation rather than mapping one label per data row.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# dnorm() evaluates density on a plotting grid; shading identifies the upper-tail event.
# The plotted range is finite although the normal distribution extends indefinitely.
curve_data <- data.frame(Miles = sort(unique(c(40000,
  seq(mu - 4 * sigma, mu + 4 * sigma, length.out = 1001)))))
```

```r
curve_data$Density <- dnorm(curve_data$Miles, mean = mu, sd = sigma)
ggplot(curve_data, aes(Miles, Density)) +
  geom_area(data = subset(curve_data, Miles >= 40000), fill = "#B6D3E1") +
  geom_line(color = "#356B8C", linewidth = .8) +
  geom_vline(xintercept = 40000, linetype = "dashed") +
  annotate("text", x = 46500, y = .000055, label = "Upper-tail area\nabout 0.2420") +
  labs(x = "Tire lifetime (miles)", y = "Density (per mile)",
       title = "A normal-model probability is an area")
```



Original R model plot using the Grear parameters checked on textbook p.286. It is not a histogram of raw tire lifetimes.

Invert the question: a quantile

Now the probability is given and the cutoff is unknown. A **10th percentile** c satisfies $P(X \leq c) = 0.10$.

```r
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# qnorm(p,...) returns the value whose lower-tail probability is p.
# Verify the result by putting the cutoff back into pnorm(); probability and miles are
 ↪  different outputs.
cutoff <- qnorm(.10, mean = mu, sd = sigma)
round_cutoff <- 30000
kable(data.frame(Quantity = c("10th percentile (miles)", "Probability below exact
 ↪  cutoff", "Probability below 30,000 miles"),
                 Value = c(cutoff, pnorm(cutoff, mu, sigma), pnorm(round_cutoff, mu,
                  ↪  sigma))), digits = 4)
```

Quantity	Value
10th percentile (miles)	30092.2422
Probability below exact cutoff	0.1000
Probability below 30,000 miles	0.0968

The unrounded 10th percentile is about **30,092 miles**. The book's approximately **30,100-mile** result uses a rounded z-table value; R calculates the quantile more precisely. A practical cutoff of 30,000 miles implies a lower-tail probability of about **9.68%**, under this model. A lifetime probability does not by itself determine a business policy: costs and other relevant evidence would also be needed.

The normal model permits negative values mathematically. For this particular parameter combination, the negative-lifetime tail is negligible, but a normal model is not automatically appropriate for nonnegative, strongly skewed measurements. Do not replace an empirical shape check with the claim that every numerical variable is normal.

A function guide after the concepts

The prefixes `d`, `p`, `q`, and `r` identify density or mass, cumulative probability, quantile, and random draws, respectively. Choose the output that answers the question.

Task	Discrete binomial example	Continuous normal example
Mass at a count / density at a value	`dbinom(x, size, prob)` gives $P(X = x)$	`dnorm(x, mean, sd)` gives a height
Probability up to a value	`pbinom(k, size, prob)`	`pnorm(t, mean, sd)`
Cutoff for a lower-tail probability	`qbinom(p, size, prob)`	`qnorm(p, mean, sd)`
Artificial model draws	`rbinom(n, size, prob)`	`rnorm(n, mean, sd)`

A discrete quantile is the smallest supported value whose cumulative probability reaches or exceeds the requested probability; an exact probability match may not exist. For continuous normal quantiles, the CDF equals the requested probability apart from numerical precision.

Complete Lesson 5 practice and the integrated review. Module 4 will distinguish an individual observation's distribution from the sampling distribution of a statistic; its lessons and practice are available.

Sources and limits

Anderson et al. (2020), 8th ed., §6.1, pp.275–277, and §6.2, pp.279–290; Grear setting and parameters, pp.286–288. Bounds, units, mean, SD, tail question, and guarantee percentile were checked against the local PDF. All charts are computed model illustrations. No flight records or tire measurements were invented, and no new workbook-to-CSV conversion is implied.

Module 3: Continuous Distributions — Practice

Lesson 5 | Calculate, justify, and interpret

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Preparation 1
1. Uniform areas 1
2. Normal probabilities 1
3. A cutoff rather than a probability 1
4. Draw and explain 2
5. Model limits 2

Preparation

Read Lesson 5. Use the flight and Grear settings, units, and stated model parameters from their English–Arabic introductions. No raw flight or tire observations are supplied.

1. Uniform areas

For the flight model $T \sim \text{Uniform}(120, 140)$ minutes, calculate $P(128 \leq T \leq 136)$, $P(T > 135)$, and $P(T = 130)$. State the density height and its units. Explain why "exactly 130" in a continuous model differs from a duration recorded after rounding to the nearest minute.

2. Normal probabilities

Use the Grear mean 36,500 miles and SD 5,000 miles.

1. Standardize 40,000 miles.
2. Find the probability of exceeding 40,000 miles, both directly and using a standard normal value.
3. Find the probability of a lifetime between 30,000 and 40,000 miles.
4. Explain why `dnorm(40000,...)` does not answer the upper-tail question.

3. A cutoff rather than a probability

Find the 10th-percentile lifetime. Check the result using `pnorm()`. Find the probability below 30,000 miles and explain the effect of rounding the practical cutoff down to that value. Distinguish miles from probability in the output.

4. Draw and explain

Create a normal density plot with the area above 40,000 miles shaded. Label the horizontal axis in miles and the vertical axis as density. Identify the parameters, the event, and the interpretation of the shaded area.

5. Model limits

Explain why not every numerical variable is normal, why a z-score alone does not justify a normal tail calculation, and what the unavailable raw road-test data prevent us from checking. Would the distribution of one tire's lifetime automatically equal the distribution of the mean lifetime of many tires? Explain without calculating a sampling distribution yet.

Submit calculations, the plot, and interpretations. Then complete the integrated review. Return to the lesson.

Module 3: Probability and Distributions Review

Connect the experiment, the model, and the interpretation

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2. Choose the denominator . . . 1
3. Interpret a discrete distribution . . . 2
4. Select a count model . . . 2
5. Separate area, height, and cutoff . . . 2
Review checklist . . . 2

Purpose and submission

Prepare a short Quarto report with the five sections below. Give the event in notation, justify the method and assumptions, show reproducible R code, and interpret every result with its units or reference group. Keep instructor answers separate.

These are the cases introduced in Lessons 1–5. Return to their English–Arabic introductions for the business settings; this review introduces no new case or observations.

1. Define the probability experiment

Use the Restaurant CSV, with uniform selection of one of its 300 records. Define A as a Very Good rating and B as a price from USD 20 to less than USD 30.

1. Find $P(A \cup B)$ and $P(A \cap B)$.
2. Explain why selecting uniformly among the three rating names would be a different experiment.
3. State why these probabilities do not establish the proportion in today's restaurant market.

2. Choose the denominator

1. Calculate $P(A \mid B)$ and $P(B \mid A)$; explain the difference.
2. Determine whether these events are independent in the finite selection experiment.
3. Using the two-supplier inputs from Lesson 2, find $P(S_1 \mid D)$ and distinguish it from $P(D \mid S_1)$.

3. Interpret a discrete distribution

Use the six published DiCarlo frequency cells in Lesson 3.

1. Construct the PMF and show that its masses sum to 1.
2. Find $P(X \geq 3)$, the expected daily count, and the model SD.
3. Explain why 1.50 is neither the mode nor a possible one-day count.
4. Distinguish the published aggregate frequencies from an individual-day dataset.

4. Select a count model

1. For Martin Clothing Store, justify the binomial model and find the probability of at least two purchases among the next three customers.
2. For the arrivals example, state the Poisson-process assumptions and calculate the probability of no arrivals in three minutes.
3. Explain why using mean count 10 in that three-minute calculation is incorrect.
4. Explain why simulation under either model does not verify its assumptions against real operations.

5. Separate area, height, and cutoff

1. Under the flight model, find $P(128 \leq T \leq 136)$ and explain why the density height is a different quantity.
2. Under the Grear normal model, find $P(X > 40{,}000)$ and the 10th-percentile lifetime in miles.
3. Explain why a z-score alone does not justify a normal model.
4. State which raw observations are unavailable and what that prevents you from checking.

Review checklist

Criterion	Evidence in the report
Clear experiment	Outcome, event, and reference group
Suitable model	Variable type, parameters, units, and assumptions
Correct event	Exact / at most / at least / more than translated correctly
Reproducible calculation	CSV for observations; cited constants for published model inputs
Sound interpretation	Probability versus density, expectation versus realization
Honest limits	Historical data, model assumptions, unavailable raw observations

This completes the selected Module 3 sequence. Proceed to Module 4: Sampling and estimation when its lessons are available, or return to the Module 3 overview.